

Week 5

Healthcare Data Preparation & Data Modeling

Power Query Cleaning, Star Schema Design and DAX Calculations in Power BI

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1. Task Overview:

1.1 Objectives:

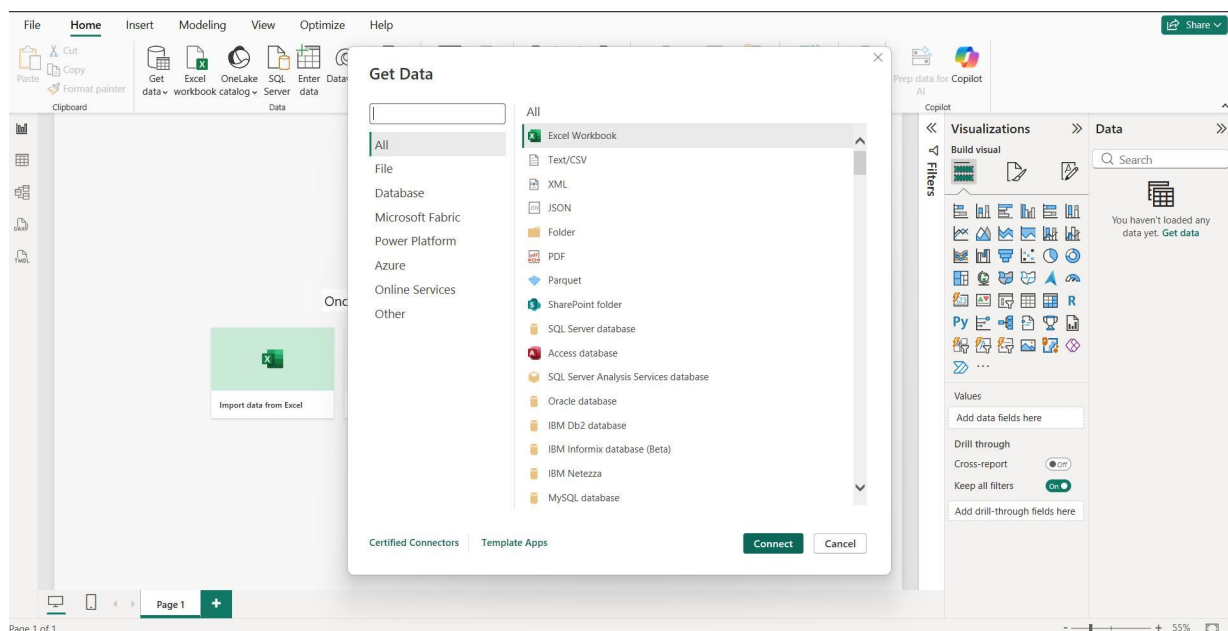
- Review patient, admission and billing data in Power Query.
- Handle missing diagnostic values and standardize text fields.
- Create Age Group categories: 0?18, 19?35, 36?60 and 60+.
- Build patient and admission dimensions with a billing fact table.
- Create DAX KPIs: Total Patients, Billing, Avg Billing, Avg Stay and Readmission Rate.

1.2 Tasks & Deliverables:

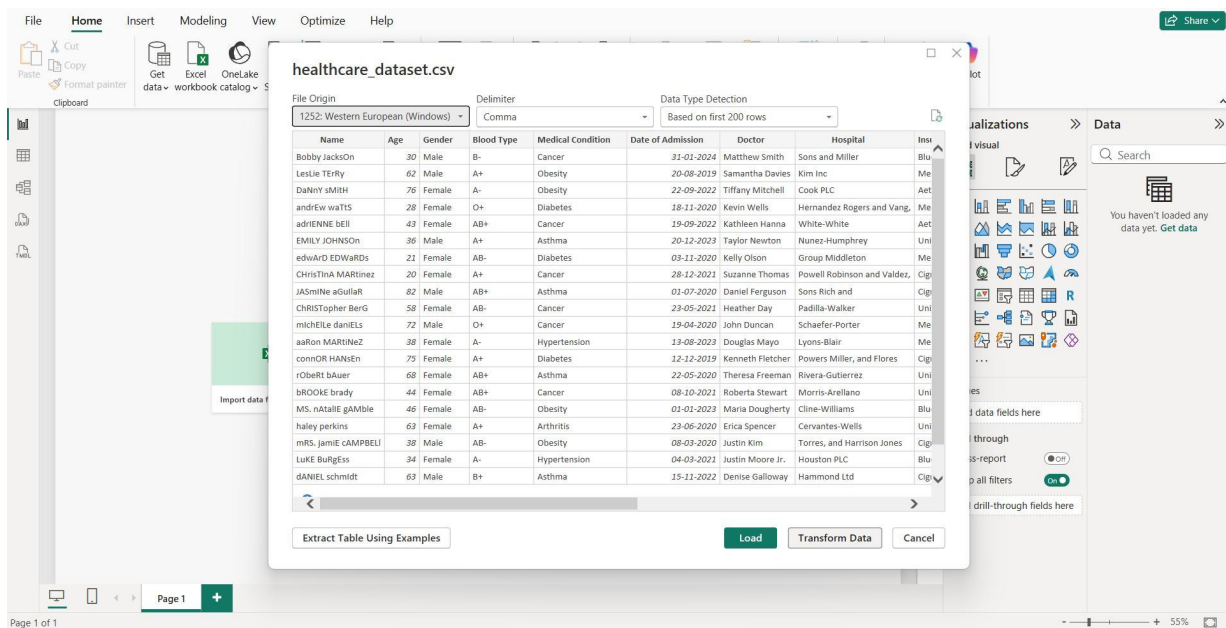
- Group patient ages into useful demographic categories.
- Standardize admission labels: Emergency, Elective and Urgent.
- Build a star schema around the Billing fact table.
- Add core KPI measures, including Readmission Rate.
- Deliver a cleaned model ready for healthcare analysis.

2. Importing the Dataset:

The healthcare_dataset.csv file was imported into Power BI Desktop using Get Data and the Text/CSV connector.

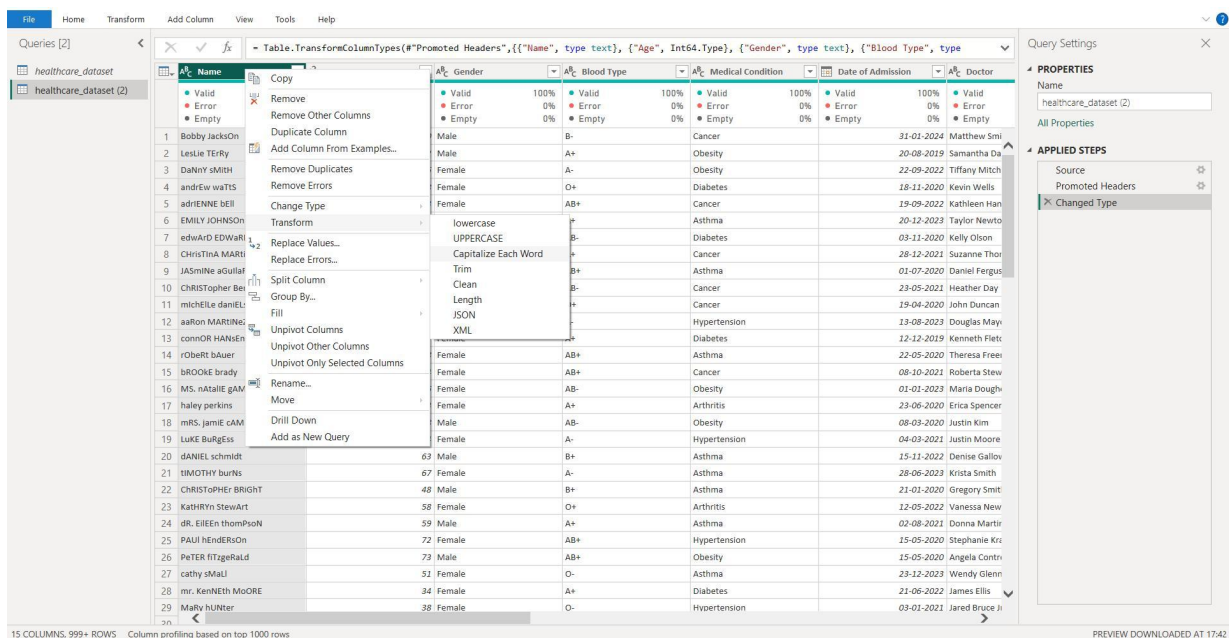


The import preview verified the file encoding as 1252: Western European, the comma separator, and automatic data-type detection from the first 200 records.



3. Standardizing Text Attributes:

The column data types were checked in Power Query. The Name field was selected and Transform > Capitalize Each Word was used so that patient names follow a uniform capitalization pattern.



4. Creating the Age Group Conditional Column:

To classify patients by age, Add Column > Conditional Column was selected and a new field called Age Group was created.

The screenshot shows the Power Query Editor interface. The main window displays a table with columns: Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, and Doctor. The 'Add Conditional Column' dialog box is open, showing the configuration for a new column named 'Age Group'. The dialog includes a table with the following columns: Column Name, Operator, Value, and Output. The 'Column Name' is 'Age Group'. The 'Operator' is 'is less than or equal to'. The 'Value' is '18'. The 'Output' is 'child'. The dialog also has an 'Add Clause' button and an 'OK' button.

Three conditions were applied to the Age field: values up to 18 were labelled Child, values up to 35 as Young, and values up to 60 as Adult. Ages above 60 were assigned to Senior.

The screenshot shows the Power Query Editor interface. The main window displays a table with columns: Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, and Doctor. The 'Add Conditional Column' dialog box is open, showing the configuration for a new column named 'Age Group'. The dialog includes a table with the following columns: Column Name, Operator, Value, and Output. The 'Column Name' is 'Age Group'. The 'Operator' is 'is less than or equal to'. The 'Value' is '18'. The 'Output' is 'child'. The dialog also has an 'Add Clause' button and an 'OK' button.

The new Age Group field appears as an additional Applied Step, and every record is assigned one of the four labels: Child, Young, Adult or Senior.

FileHomeTransformAdd ColumnViewToolsHelp

Queries [1]

fx

= Table.AddColumn("Changed Type", "Age Group", each if [Age] <= 18 then "Child" else if [Age] <= 35 then "Young" else if [Age] <= 60 then

healthcare_dataset

100%Valid0%Error0%Empty

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	Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group
1	18856.28131	328 Urgent	02-02-2024	Paracetamol	Normal	Young
2	33643.32729	265 Emergency	26-08-2019	Ibuprofen	Inconclusive	Senior
3	27955.09608	205 Emergency	07-10-2022	Aspirin	Normal	Senior
4	37909.78241	450 Elective	18-12-2020	Ibuprofen	Abnormal	Young
5	14238.31781	458 Urgent	09-10-2022	Penicillin	Abnormal	Adult
6	48145.11095	389 Urgent	24-12-2023	Ibuprofen	Normal	Adult
7	19580.87234	389 Emergency	15-11-2020	Paracetamol	Inconclusive	Young
8	45820.46272	277 Emergency	07-01-2022	Paracetamol	Inconclusive	Young
9	50119.22279	316 Elective	14-07-2020	Aspirin	Abnormal	Senior
10	19784.63106	249 Elective	22-06-2021	Paracetamol	Inconclusive	Adult
11	12576.79561	394 Urgent	22-04-2020	Paracetamol	Normal	Senior
12	7999.58688	288 Urgent	05-09-2023	Lipitor	Inconclusive	Adult
13	43282.28336	134 Emergency	28-12-2019	Penicillin	Abnormal	Senior
14	33207.70663	309 Urgent	19-06-2020	Lipitor	Normal	Senior
15	40701.59923	182 Urgent	13-10-2021	Paracetamol	Normal	Adult
16	12263.35743	465 Elective	11-01-2023	Aspirin	Inconclusive	Adult
17	24499.8479	114 Elective	14-07-2020	Paracetamol	Normal	Senior
18	17440.46544	449 Urgent	02-04-2020	Paracetamol	Abnormal	Adult
19	18843.02302	260 Elective	14-03-2021	Aspirin	Abnormal	Young
20	23762.20358	465 Elective	22-11-2022	Penicillin	Normal	Senior
21	42.51458855	115 Elective	02-07-2023	Aspirin	Normal	Senior
22	17695.91162	295 Urgent	09-02-2020	Lipitor	Normal	Adult
23	5998.102908	327 Urgent	10-06-2022	Lipitor	Inconclusive	Adult
24	25250.05243	119 Urgent	12-08-2021	Lipitor	Inconclusive	Adult
25	33211.29542	109 Emergency	08-06-2020	Paracetamol	Inconclusive	Senior
26	19746.83201	162 Urgent	20-05-2020	Aspirin	Abnormal	Senior
27	26786.52956	401 Elective	19-01-2024	Ibuprofen	Normal	Adult
28	18834.80134	157 Emergency	30-06-2022	Lipitor	Abnormal	Young
29	32643.29935	223 Emergency	16-01-2021	Penicillin	Normal	Adult

16 COLUMNS, 999+ ROWSColumn profiling based on top 1000 rows

Query Settings

healthcare_dataset

APPLIED STEPS

SourcePromoted HeadersChanged TypeAdded Conditional Column

5. Adding Surrogate Keys:

For connecting the tables later, two unique index-based keys were generated. Add Column > Index Column > From 1 was used to create a sequential number for each record.

FileHomeTransformAdd ColumnViewToolsHelp

Column From Custom Examples * Column

Conditional Column

Index Column

From 1

From Text

From Number

From Date & Time

Create a new column with an index starting at 1.

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

0%Valid0%Error0%Empty

1	Bobby	30 Male	B-	Cancer		31-01-2024 Matthew Smt
2	Leslie Terry	62 Male	A+	Obesity		20-08-2019 Samantha Da
3	Danny Smith	76 Female	A-	Obesity		22-09-2022 Tiffany Mitch
4	Andrew Watts	28 Female	O+	Diabetes		18-11-2020 Kevin Wells
5	Adrienne Bell	43 Female	AB+	Cancer		19-09-2022 Kathleen Han
6	Emily Johnson	36 Male	A+	Asthma		20-12-2023 Taylor Newto
7	Edward Edwards	21 Female	AB-	Diabetes		03-11-2020 Kelly Olson
8	Christina Martinez	20 Female	A+	Cancer		28-12-2021 Suzanne Thor
9	Jasmine Aguilar	82 Male	AB+	Asthma		01-07-2020 Daniel Fergus
10	Christopher Berg	58 Female	AB-	Cancer		23-05-2021 Heather Day
11	Michelle Daniels	72 Male	O+	Cancer		19-04-2020 John Duncan
12	Aaron Martinez	38 Female	A-	Hypertension		13-08-2023 Douglas May
13	Connor Hansen	75 Female	A+	Diabetes		12-12-2019 Kenneth Flet
14	Robert Bauer	68 Female	AB+	Asthma		22-05-2020 Theresa Free
15	Brooke Brady	44 Female	AB+	Cancer		08-10-2021 Roberta Stew
16	Ms. Natalie Gamble	46 Female	AB-	Obesity		01-01-2023 Maria Dough
17	Haley Perkins	63 Female	A+	Arthritis		29-06-2020 Erica Spence
18	Mrs. Jamie Campbell	38 Male	AB-	Obesity		08-03-2020 Justin Kim
19	Luke Burgess	34 Female	A-	Hypertension		04-03-2021 Justin Moore
20	Daniel Schmidt	63 Male	B+	Asthma		15-11-2022 Denise Gallov
21	Timothy Burns	67 Female	A-	Asthma		28-06-2023 Krista Smith
22	Christopher Bright	48 Male	B+	Asthma		21-01-2020 Gregory Smit
23	Kathryn Stewart	58 Female	O+	Arthritis		12-05-2022 Vanessa New
24	Dr. Eileen Thompson	59 Male	A+	Asthma		02-08-2021 Donna Marti
25	Paul Henderson	72 Female	AB+	Hypertension		15-05-2020 Stephanie Kr
26	Peter Fitzgerald	73 Male	AB+	Obesity		15-05-2020 Angelia Contr
27	Cathy Small	51 Female	O-	Asthma		23-12-2023 Wendy Glenn
28	Mr. Kenneth Moore	34 Female	A+	Diabetes		21-06-2022 James Ellis
29	Mary Hunter	38 Female	O-	Hypertension		03-01-2021 Jared Bruce Ji

15 COLUMNS, 999+ ROWSColumn profiling based on top 1000 rows

healthcare_dataset (2)

APPLIED STEPS

SourcePromoted HeadersChanged TypeCapitalized Each Word

A second index was created using the same method. Both index fields were then renamed through the Rename option.

The screenshot shows a data table with 17 columns and 999+ rows. The columns are: Admission Type, Discharge Date, Medication, Test Results, and Index. A context menu is open over the 'Index' column, showing options like Copy, Remove, Duplicate Column, Add Column From Examples..., Remove Duplicates, Remove Errors, Change Type, Transform, Replace Values..., Replace Errors..., Group By..., Fill, Unpivot Columns, Unpivot Other Columns, Unpivot Only Selected Columns, Rename..., Move, Drill Down, and Add as New Query. The 'Rename...' option is highlighted. The 'Query Settings' panel on the right shows the 'PROPERTIES' tab with the name 'healthcare_dataset (2)' and the 'APPLIED STEPS' tab with 'Added Index1'.

	Admission Type	Discharge Date	Medication	Test Results	Index
1	Urgent	02-02-2024	Paracetamol	Normal	
2	Emergency	26-08-2019	Ibuprofen	Inconclusive	
3	Emergency	07-10-2022	Aspirin	Normal	
4	Elective	18-12-2020	Ibuprofen	Abnormal	
5	Urgent	09-10-2022	Penicillin	Abnormal	
6	Urgent	24-12-2023	Ibuprofen	Normal	
7	Emergency	15-11-2020	Paracetamol	Inconclusive	
8	Emergency	07-01-2022	Paracetamol	Inconclusive	
9	Elective	14-07-2020	Aspirin	Abnormal	
10	Elective	22-06-2021	Paracetamol	Inconclusive	
11	Urgent	22-04-2020	Paracetamol	Normal	
12	Urgent	05-09-2023	Lipitor	Inconclusive	
13	Emergency	28-12-2019	Penicillin	Abnormal	
14	Urgent	19-06-2020	Lipitor	Normal	
15	Urgent	13-10-2021	Paracetamol	Normal	
16	Elective	11-01-2023	Aspirin	Inconclusive	
17	Elective	14-07-2020	Paracetamol	Normal	
18	Urgent	02-04-2020	Paracetamol	Abnormal	
19	Elective	14-03-2021	Aspirin	Abnormal	
20	Elective	22-11-2022	Penicillin	Normal	
21	Elective	02-07-2023	Aspirin	Normal	
22	Urgent	09-02-2020	Lipitor	Normal	
23	Urgent	10-06-2022	Lipitor	Inconclusive	
24	Urgent	12-08-2021	Lipitor	Inconclusive	
25	Emergency	08-06-2020	Paracetamol	Inconclusive	
26	Urgent	20-05-2020	Aspirin	Abnormal	
27	Elective	19-01-2024	Ibuprofen	Normal	
28	Emergency	30-06-2022	Lipitor	Abnormal	
29	Emergency	16-01-2021	Penicillin	Normal	

The two indexes were given the names Patient ID and Admission ID. These identifiers can therefore be reused when creating the patient, admission and billing tables.

The screenshot shows the same data table as before, but the 'Index' column has been renamed to 'Admission ID'. The 'Query Settings' panel on the right shows the 'PROPERTIES' tab with the name 'healthcare_dataset (2)' and the 'APPLIED STEPS' tab with 'Renamed Columns'.

	Admission Type	Discharge Date	Medication	Test Results	Patient ID	Admission ID
1	Urgent	02-02-2024	Paracetamol	Normal	1	1
2	Emergency	26-08-2019	Ibuprofen	Inconclusive	2	2
3	Emergency	07-10-2022	Aspirin	Normal	3	3
4	Elective	18-12-2020	Ibuprofen	Abnormal	4	4
5	Urgent	09-10-2022	Penicillin	Abnormal	5	5
6	Urgent	24-12-2023	Ibuprofen	Normal	6	6
7	Emergency	15-11-2020	Paracetamol	Inconclusive	7	7
8	Emergency	07-01-2022	Paracetamol	Inconclusive	8	8
9	Elective	14-07-2020	Aspirin	Abnormal	9	9
10	Elective	22-06-2021	Paracetamol	Inconclusive	10	10
11	Urgent	22-04-2020	Paracetamol	Normal	11	11
12	Urgent	05-09-2023	Lipitor	Inconclusive	12	12
13	Emergency	28-12-2019	Penicillin	Abnormal	13	13
14	Urgent	19-06-2020	Lipitor	Normal	14	14
15	Urgent	13-10-2021	Paracetamol	Normal	15	15
16	Elective	11-01-2023	Aspirin	Inconclusive	16	16
17	Elective	14-07-2020	Paracetamol	Normal	17	17
18	Urgent	02-04-2020	Paracetamol	Abnormal	18	18
19	Elective	14-03-2021	Aspirin	Abnormal	19	19
20	Elective	22-11-2022	Penicillin	Normal	20	20
21	Elective	02-07-2023	Aspirin	Normal	21	21
22	Urgent	09-02-2020	Lipitor	Normal	22	22
23	Urgent	10-06-2022	Lipitor	Inconclusive	23	23
24	Urgent	12-08-2021	Lipitor	Inconclusive	24	24
25	Emergency	08-06-2020	Paracetamol	Inconclusive	25	25
26	Urgent	20-05-2020	Aspirin	Abnormal	26	26
27	Elective	19-01-2024	Ibuprofen	Normal	27	27
28	Emergency	30-06-2022	Lipitor	Abnormal	28	28
29	Emergency	16-01-2021	Penicillin	Normal	29	29

After adding Age Group, Patient ID and Admission ID, the main healthcare_dataset query contains 18 columns and is used as the base for the remaining queries.

Query: `Table.RenameColumns(#"Added Index1",{{"Index", "Admission ID"}, {"Index.1", "Patient ID"}})`

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1 Bobby Jackson	30	Male	B-	Cancer	31-01-2024	Matthew Smi
2 Leslie Terry	62	Male	A+	Obesity	20-08-2019	Samantha Da
3 Danny Smith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitch
4 Andrew Watts	28	Female	O+	Diabetes	18-11-2020	Kevin Wells
5 Adrienne Bell	43	Female	AB+	Cancer	19-09-2022	Kathleen Han
6 Emily Johnson	36	Male	A+	Asthma	20-12-2023	Taylor Newto
7 Edward Edwards	21	Female	AB-	Diabetes	03-12-2020	Kelly Olson
8 Christina Martinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thor
9 Jasmine Aguilar	82	Male	AB+	Asthma	01-07-2020	Daniel Fergus
10 Christopher Berg	58	Female	AB-	Cancer	23-05-2021	Heather Day
11 Michelle Daniels	72	Male	O+	Cancer	19-04-2020	John Duncan
12 Aaron Martinez	38	Female	A-	Hypertension	13-08-2023	Douglas May
13 Connor Hansen	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletc
14 Robert Bauer	68	Female	AB+	Asthma	22-05-2020	Theresa Freee
15 Brooke Brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stew
16 Ms. Natalie Gamble	46	Female	AB-	Obesity	01-01-2023	Maria Dough
17 Haley Perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer
18 Mrs. Jamie Campbell	38	Male	AB-	Obesity	08-09-2020	Justin Kim
19 Luke Burgess	34	Female	A-	Hypertension	04-03-2021	Justin Moore
20 Daniel Schmidt	63	Male	B+	Asthma	15-11-2022	Denise Gallow
21 Timothy Burns	67	Female	A-	Asthma	28-06-2023	Krista Smith
22 Christopher Bright	48	Male	B+	Asthma	21-01-2020	Gregory Smit
23 Kathryn Stewart	58	Female	O+	Arthritis	12-05-2022	Vanessa New
24 Dr. Eileen Thompson	59	Male	A+	Arthritis	02-08-2021	Donna Martir
25 Paul Henderson	72	Female	AB+	Hypertension	15-05-2020	Stephanie Kri
26 Peter Fitzgerald	73	Male	AB+	Obesity	15-05-2020	Angela Contr
27 Cathy Small	51	Female	O-	Asthma	23-12-2023	Wendy Glenn
28 Mr. Kenneth Moore	34	Female	A+	Diabetes	21-06-2022	James Ellis
29 Mary Hunter	38	Female	O-	Hypertension	03-01-2021	Jared Bruce Ji

18 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 19:41

6. Building the Dim_Patient Dimension Table:

A reference query named Dim_Patient was created from healthcare_dataset. Fields that were not patient descriptors, such as Medical Condition, were removed from this query.

Query: `healthcare_dataset`

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1 Bobby Jackson	30	Male	B-	Cancer	31-01-2024	Matthew Smi
2 Leslie Terry	62	Male	A+	Obesity	20-08-2019	Samantha Da
3 Danny Smith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitch
4 Andrew Watts	28	Female	O+	Diabetes	18-11-2020	Kevin Wells
5 Adrienne Bell	43	Female	AB+	Cancer	19-09-2022	Kathleen Han
6 Emily Johnson	36	Male	A+	Asthma	20-12-2023	Taylor Newto
7 Edward Edwards	21	Female	AB-	Diabetes	03-12-2020	Kelly Olson
8 Christina Martinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thor
9 Jasmine Aguilar	82	Male	AB+	Asthma	01-07-2020	Daniel Fergus
10 Christopher Berg	58	Female	AB-	Cancer	23-05-2021	Heather Day
11 Michelle Daniels	72	Male	O+	Cancer	19-04-2020	John Duncan
12 Aaron Martinez	38	Female	A-	Hypertension	13-08-2023	Douglas May
13 Connor Hansen	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletc
14 Robert Bauer	68	Female	AB+	Asthma	22-05-2020	Theresa Freee
15 Brooke Brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stew
16 Ms. Natalie Gamble	46	Female	AB-	Obesity	01-01-2023	Maria Dough
17 Haley Perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer
18 Mrs. Jamie Campbell	38	Male	AB-	Obesity	08-09-2020	Justin Kim
19 Luke Burgess	34	Female	A-	Hypertension	04-03-2021	Justin Moore
20 Daniel Schmidt	63	Male	B+	Asthma	15-11-2022	Denise Gallow
21 Timothy Burns	67	Female	A-	Asthma	28-06-2023	Krista Smith
22 Christopher Bright	48	Male	B+	Asthma	21-01-2020	Gregory Smit
23 Kathryn Stewart	58	Female	O+	Arthritis	12-05-2022	Vanessa New
24 Dr. Eileen Thompson	59	Male	A+	Arthritis	02-08-2021	Donna Martir
25 Paul Henderson	72	Female	AB+	Hypertension	15-05-2020	Stephanie Kri
26 Peter Fitzgerald	73	Male	AB+	Obesity	15-05-2020	Angela Contr
27 Cathy Small	51	Female	O-	Asthma	23-12-2023	Wendy Glenn
28 Mr. Kenneth Moore	34	Female	A+	Diabetes	21-06-2022	James Ellis
29 Mary Hunter	38	Female	O-	Hypertension	03-01-2021	Jared Bruce Ji

16 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 19:39

After removing unnecessary fields, Dim_Patient retains six columns: Name, Age, Gender, Blood Type, Age Group and Patient ID.

	Name	Age	Gender	Blood Type	Age Group	Patient ID
1	Bobby Jackson	30	Male	B-	Young	1
2	Leslie Terry	62	Male	A+	Senior	2
3	Danny Smith	76	Female	A-	Senior	3
4	Andrew Watts	28	Female	O+	Young	4
5	Adrienne Bell	43	Female	AB+	Adult	5
6	Emily Johnson	36	Male	A+	Adult	6
7	Edward Edwards	21	Female	AB-	Young	7
8	Christina Martinez	20	Female	A+	Young	8
9	Jasmine Aguilar	82	Male	AB+	Senior	9
10	Christopher Berg	58	Female	AB-	Adult	10
11	Michelle Daniels	72	Male	O+	Senior	11
12	Aaron Martinez	38	Female	A-	Adult	12
13	Connor Hansen	75	Female	A+	Senior	13
14	Robert Bauer	68	Female	AB+	Senior	14
15	Brooke Brady	44	Female	AB+	Adult	15
16	Ms. Natalie Gamble	46	Female	AB-	Adult	16
17	Haley Perkins	63	Female	A+	Senior	17
18	Mrs. Jamie Campbell	38	Male	AB-	Adult	18
19	Luke Burgess	34	Female	A-	Young	19
20	Daniel Schmidt	64	Male	B+	Senior	20
21	Timothy Burns	67	Female	A-	Senior	21
22	Christopher Bright	48	Male	B+	Adult	22
23	Kathryn Stewart	58	Female	O+	Adult	23
24	Dr. Eileen Thompson	59	Male	A+	Adult	24
25	Paul Henderson	72	Female	AB+	Senior	25
26	Peter Fitzgerald	73	Male	AB+	Senior	26
27	Cathy Small	51	Female	O-	Adult	27
28	Mr. Kenneth Moore	34	Female	A+	Young	28
29	Mary Hunter	38	Female	O-	Adult	29
30	Jacob Olsen	43	Female	DA	Adult	30

7. Building the Billing Table:

The Billing query was derived from the main query by deleting patient and admission description fields. It retains Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient ID and Admission ID, allowing each billing entry to be linked back to the relevant records.

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Admission ID	Patient ID
1	31-01-2024	Blue Cross	18856.28131	02-02-2024	1	1
2	20-08-2019	Medicare	33643.32729	26-08-2019	2	2
3	22-09-2022	Aetna	27955.09608	07-10-2022	3	3
4	18-11-2020	Medicare	37909.78241	18-12-2020	4	4
5	19-09-2022	Aetna	14238.31781	09-10-2022	5	5
6	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023	6	6
7	03-11-2020	Medicare	19580.87234	15-11-2020	7	7
8	28-12-2021	Cigna	45820.46272	07-01-2022	8	8
9	01-07-2020	Cigna	50119.22279	14-07-2020	9	9
10	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021	10	10
11	19-04-2020	Medicare	12576.79561	22-04-2020	11	11
12	13-08-2023	Medicare	7999.58688	05-09-2023	12	12
13	12-12-2019	Cigna	43282.28336	28-12-2019	13	13
14	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020	14	14
15	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021	15	15
16	01-01-2023	Blue Cross	12263.35743	11-01-2023	16	16
17	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020	17	17
18	08-03-2020	Cigna	17440.46544	02-04-2020	18	18
19	04-03-2021	Blue Cross	18843.02302	14-03-2021	19	19
20	15-11-2022	Cigna	23762.20358	22-11-2022	20	20
21	28-06-2023	Blue Cross	42.51458855	02-07-2023	21	21
22	21-01-2020	Aetna	17695.91162	09-02-2020	22	22
23	12-05-2022	Aetna	5998.102908	10-06-2022	23	23
24	02-08-2021	Aetna	25250.05243	12-08-2021	24	24
25	15-05-2020	Medicare	33211.29542	08-06-2020	25	25
26	15-05-2020	Medicare	19746.83201	20-05-2020	26	26
27	23-12-2022	Blue Cross	26786.52956	19-01-2024	27	27
28	21-06-2022	UnitedHealthcare	18834.80134	30-06-2022	28	28
29	03-01-2021	Cigna	32643.29935	16-01-2021	29	29
30	02-10-2022	Aetna	67673.01064	13-10-2022	30	30

8. Building the Dim_Admission Dimension Table:

A separate Dim_Admission query was also referenced from the main dataset. It contains Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number and Test Results. The Admission Type values remain Emergency, Elective and Urgent, while Admission ID serves as the primary key.

9. Applying Changes and Loading the Model:

After completing the Power Query transformations, Close & Apply was selected to load the prepared tables into Power BI Desktop.

[illegible]

10. Final Data Model:

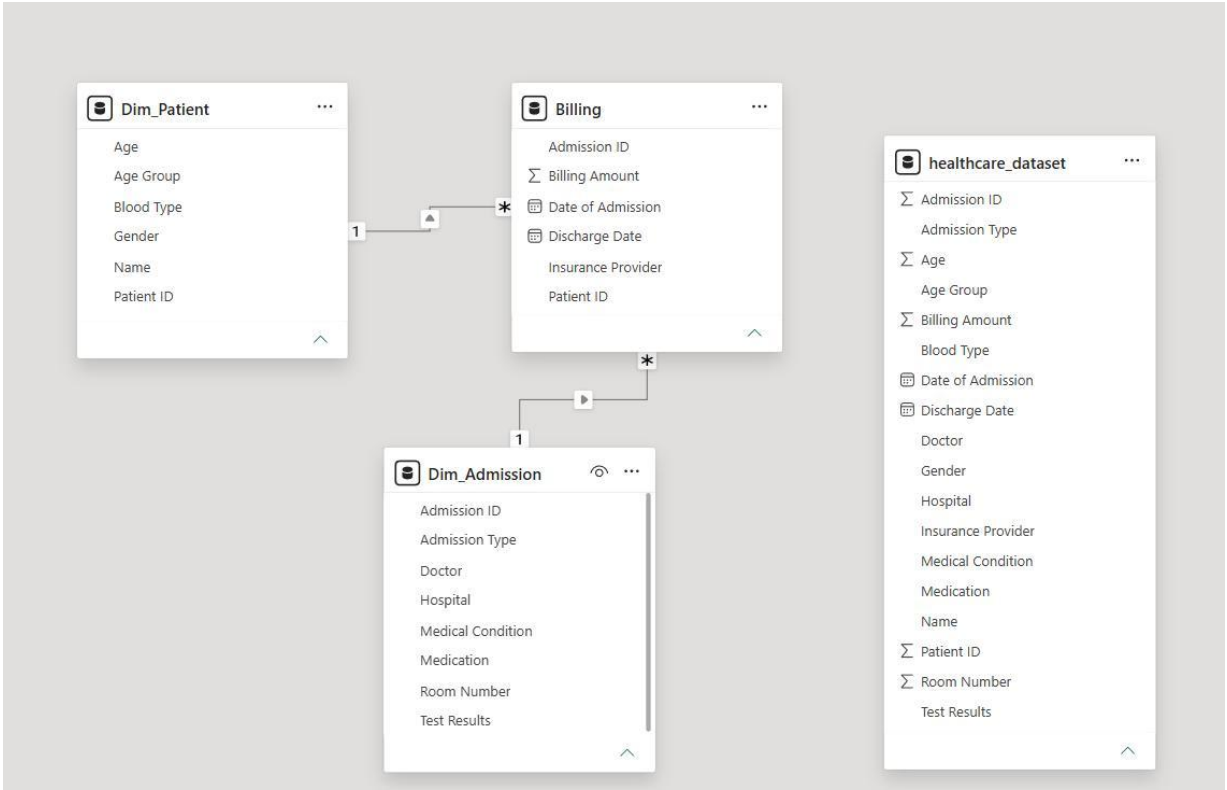
The Model view contains Dim_Patient, Dim_Admission, Billing and the original healthcare_dataset query. Since Billing contains Patient ID and Admission ID matching the keys in both dimension tables, it functions as the central fact table in the star schema.

The screenshot displays the Microsoft Fabric Data Explorer interface. The top navigation bar includes 'File', 'Home', and 'Help' tabs. Below the navigation bar, there are icons for various data sources and actions, such as 'Get data', 'Excel', 'OneLake', 'SQL Server', 'Enter data', 'Database', 'Recent sources', 'Transform Refresh', 'Manage relationships', 'New measure column', 'New table', 'Calculation group', 'New parameter', 'Parameters', 'Manage roles', 'View as', 'Q&A setup', 'Language', 'Linguistic schema', 'Sensitivity', and 'Publish'.

The main workspace shows three data tables: 'Dim_Patient', 'Billing', and 'Dim_Admission'. Each table is represented by a card with a list of fields. The 'Dim_Patient' table has fields: Age, Age Group, Blood Type, Gender, Name, and Patient ID. The 'Billing' table has fields: Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, and Patient ID. The 'Dim_Admission' table has fields: Admission ID, Admission Type, Doctor, Hospital, Insurance Provider, Medical Condition, Medication, Room Number, and Test Results.

On the right side, there is a 'Properties' pane with a 'Cards' section. It includes options to 'Show the database in the header when applicable' (set to 'No'), 'Show related fields when card is collapsed' (set to 'Yes'), and 'Pin related fields to top of card' (set to 'No'). Below the 'Properties' pane is a 'Data' pane with a search bar and a list of tables: 'Billing', 'Dim_Admission', 'Dim_Patient', and 'healthcare_dataset'.

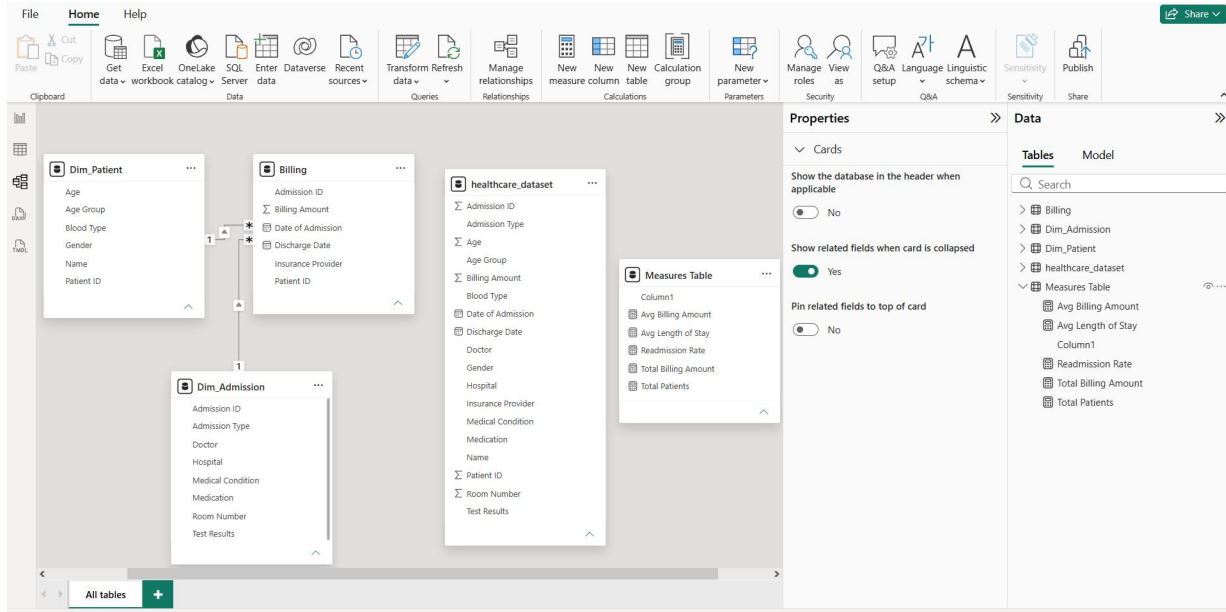
Relationships were created between Dim_Patient and Billing using Patient ID, and between Dim_Admission and Billing using Admission ID. Both relationships are one-to-many, with filtering flowing from the dimension tables toward the Billing fact table. This completes the star-schema structure.



11. Creating the Measures Table:

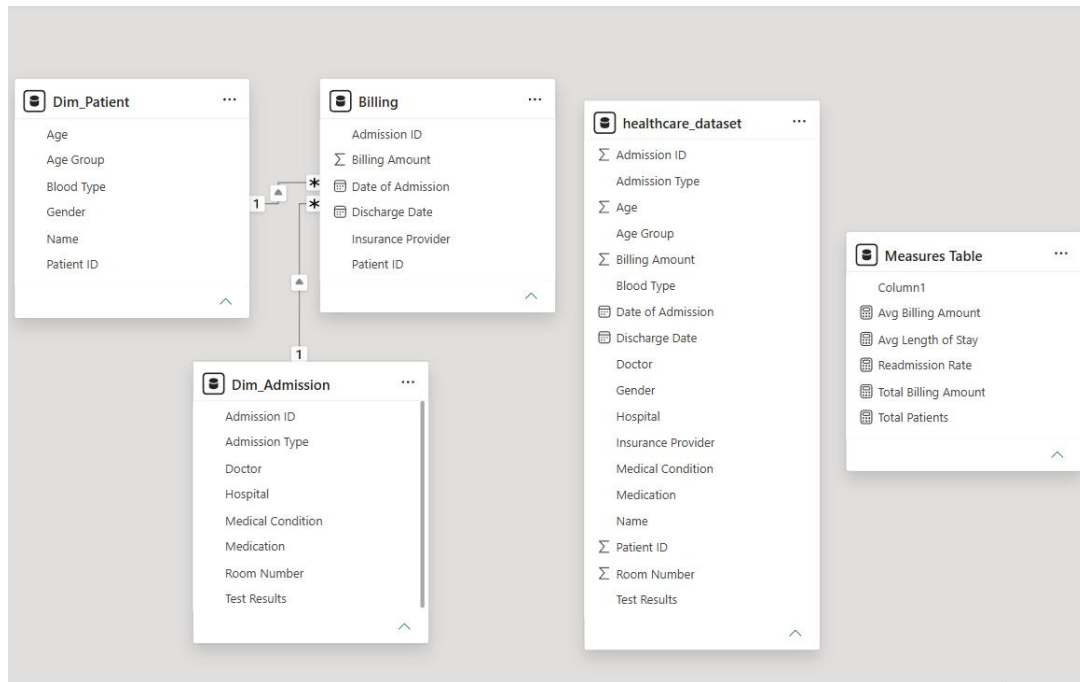
To organize the calculations in one place, a separate table named Measures Table was created for the DAX measures rather than storing them within the data tables.

Five measures were added: Total Patients, Total Billing Amount, Avg Billing Amount, Avg Length of Stay and Readmission Rate. The Model view also shows the Measures Table alongside Billing, Dim_Patient, Dim_Admission and healthcare_dataset.



12. Measures Table in the Data Model:

The Model view shows the Measures Table placed beside Dim_Patient, Billing, Dim_Admission and healthcare_dataset. Since it contains only measures, it has no relationship with the other tables, while the dimension-to-Billing relationships remain unchanged.



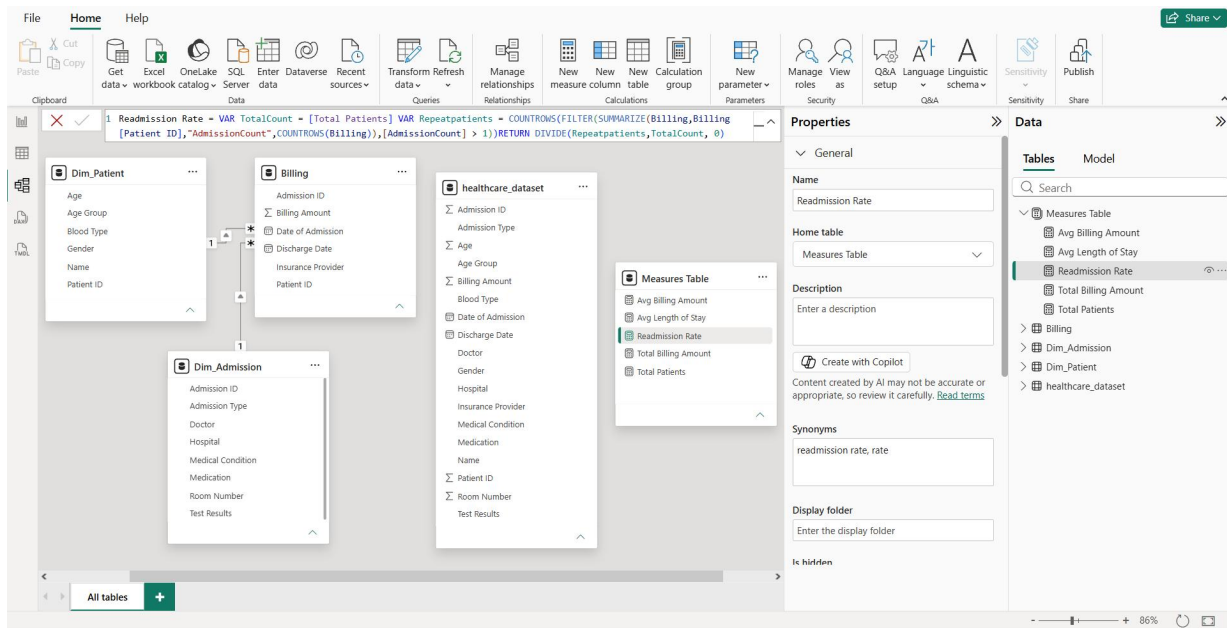
13. Writing the Readmission Rate Measure:

The Readmission Rate calculation was entered in the Power BI formula bar. It stores the Total Patients value in TotalCount, groups Billing records by Patient ID, counts admissions for each patient, and selects patients with more than one admission. The number of repeat patients is then divided by the total patient count with DIVIDE, returning zero when the denominator is zero.

```
Readmission Rate =  
VAR TotalCount = [Total Patients]  
VAR Repeatpatients =  
    COUNTROWS(  
        FILTER(  
            SUMMARIZE(  
                Billing,  
                Billing[Patient ID],  
                "AdmissionCount", COUNTROWS(Billing)  
            ),  
            [AdmissionCount] > 1  
        )  
    )  
RETURN  
    DIVIDE(Repeatpatients, TotalCount, 0)
```

In the Properties pane, the measure is named Readmission Rate, stored under Measures Table, and given the synonyms ?readmission rate, rate? for easier discovery through Q&A.

The placeholder Column1 was removed, leaving only the five required measures: Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount and Total Patients.



14. Conclusion:

The healthcare dataset was prepared and reorganized using Power Query. Patient names were standardized, an Age Group field was created, and Patient ID and Admission ID keys were generated. The data was then separated into Dim_Patient, Dim_Admission and Billing tables and loaded into Power BI Desktop. One-to-many links from the two dimension tables to Billing establish the star schema. Finally, a Measures Table was created with five DAX calculations, including the Readmission Rate, making the model suitable for healthcare analytics.